

Group Project

Applied Computer Vision (2-Week Project)

Overview

In this project, you will work in teams of **four students** to design, train, and evaluate a **computer vision model** using transfer learning.

Your task is to choose a dataset and problem, adapt a pretrained deep network, fine-tune it, and analyze the results.

This is a short but complete project that combines model development, experimentation, and communication of your results exactly like a small-scale industry or research project.

Goals

- Practice **transfer learning** and **fine-tuning** on a real dataset.
- Gain experience with modern architectures such as **ResNet**, **MobileNet**, **EfficientNet**, or **Vision Transformer (ViT)**.
- Learn to interpret training curves, analyze overfitting, and compare models.
- Build a reproducible project with proper documentation and presentation.

Suggested Project Ideas

You can pick one of the following or propose your own idea (subject to approval). All of these can be implemented in Colab within two weeks using public datasets.

1. Image Classification

- Cat vs. Dog image classifier (Kaggle dataset).
- Food-101 – identify different food types.
- Rock-Paper-Scissors hand gesture recognition.
- Flower species classification (TensorFlow Flowers dataset).
- Emotion recognition from facial expressions (FER2013).

2. Object Detection

- Detecting traffic signs (GTSRB dataset).

- Helmet detection for construction safety.
- Counting fruits in images (apple, banana, orange).
- Mask vs. no-mask face detection (Kaggle).
- Waste detection – classify and localize plastic vs. paper items.

3. Specialized or Creative Tasks

- Artwork style classification (Monet vs. Van Gogh vs. Picasso).
- Bird species classification using fine-tuned EfficientNet.
- Skin lesion classification (HAM10000 dataset).
- AI for wildlife conservation – detect animals in camera-trap images.
- Visual defect detection on products or manufactured parts.

4. Your Own Dataset

Collect a small dataset of at least 50–100 - label it, and fine-tune a model on it.

Project Workflow

Your project must include the following stages:

1. Dataset and Preprocessing

- Choose your dataset and describe it (size, classes, balance).
- Split into train, validation, and test sets.
- Apply data augmentations such as flips, rotations, and brightness changes.

2. Model and Training

- Use a pretrained backbone (e.g., ResNet50, MobileNetV2, ViT).
- Freeze/unfreeze layers to apply transfer learning.
- Train for at least 30 epochs.
- Experiment with at least one parameter (learning rate, dropout, or batch size).

3. Evaluation

- Plot training and validation accuracy/loss.

- Compute test accuracy and inference time.
- Display example predictions.

4. Analysis

- Summarize key findings (e.g., trade-off between speed and accuracy).
- Discuss overfitting or underfitting signs.
- Suggest improvements or alternative architectures.

Deliverables

Each team must submit and present:

1. GitHub Repository

- Organized code with clear structure and comments.
- A complete **README.md** including:
 - Project summary and team members
 - Dataset description
 - Model and methods
 - Results and plots
 - Setup instructions (requirements.txt, Colab link, etc.)

2. Short Report (2–3 pages, PDF)

- Introduction and motivation
- Dataset and preprocessing
- Model and training description
- Results and discussion (include plots and metrics)
- Conclusion and future work

3. Class Presentation (7-10 minutes)

- Slides with visuals (graphs, images, examples).
- Clear explanation of your approach and results.

- Each team member speaks for part of the presentation.

Evaluation (100 Points Total)

Category	Description	Points
Code & Implementation	Clean, working, reproducible code	30
Experiments & Results	Proper fine-tuning, augmentations, plots	25
Analysis & Understanding	Insights, interpretation, reflection	20
Presentation	Clarity, visuals, teamwork	15
Documentation	GitHub README + short report quality	10

Timeline

- **Week 1:** Choose dataset, prepare preprocessing, set up training.
- **Week 2:** Fine-tune, evaluate, write report, and prepare presentation.
- **Submission:** GitHub link + PDF report + slides (end of Week 2).
- **Presentation:** During final class session.